<div align="center"> <h1> 🛡️ ShieldGig</h1> <h3>Real-Time Income Protection Engine for India's Gig Delivery Workers</h3>

  🏆 Guidewire DEVTrails 2026 — Phase 1 Submission
  👥 Team: [Team Name] | 🎯 Persona: Food Delivery Partners (Zomato / Swiggy) </div>

> *"When it rains heavily, I can't deliver. When there's a curfew, I can't deliver. When the app crashes, I can't deliver. Those days, I earn zero rupees — but my rent doesn't know that."* — **Ravi, 26, Zomato delivery partner, Chennai**

⚡ How ShieldGig Works — 15-Second View

```
1. Rain detected in Ravi's zone  →  IMD + OpenWeatherMap confirm threshold
2. Shift window check passes     →  disruption falls within Ravi's working hours
3. Fraud check in < 2 seconds   →  FRS score computed across 6 signal layers
4. ₹225 credited immediately    →  ₹100/hr × 3 hrs × 0.75 coverage factor
```

No forms. No adjusters. No claim ever filed by the worker — for weather events.

🔴 The Problem

India has **7.7 million** gig delivery workers earning ₹4,000–₹6,000 per week with no paid leave, no sick days, and no safety net. Chennai alone sees **~80 rain days per year.** On a heavy rain day, a rider loses ₹400–₹600. Cyclone Michaung wiped out 3–4 days of income per worker with no recourse.

Every existing insurance product covers accidents, hospitalization, and death — events that happen rarely. Not one covers the income disruption that happens 80+ days a year.

ShieldGig fixes the right problem.

💡 What ShieldGig Is

ShieldGig is **not an insurance company.** It is an **underwriting intelligence engine** that enables licensed insurers to profitably serve gig workers — a segment traditional insurance has never been able to reach.

🏢 Insurance Partner Model

Role	Entity
Risk Underwriter	Licensed insurer — ICICI Lombard / HDFC ERGO
Trigger + Intelligence Engine	ShieldGig
Policy Administration	Guidewire PolicyCenter API
Claims Automation	Guidewire ClaimCenter API
Premium Billing	Guidewire BillingCenter API
Distribution	Zomato / Swiggy platform integration

⚙️ Guidewire Integration

PolicyCenter

- Weekly policy creation every Monday via PolicyCenter API
- ISS score + city risk profile passed as risk attributes for premium computation
- Policy status synced back to ShieldGig in real time

ClaimCenter

- On parametric trigger: ShieldGig pushes a structured, pre-validated claim payload
- Fraud Risk Score attached — ClaimCenter routes CLEAN to auto-approval, FLAGGED to human queue
- On manual claim: worker-submitted proof package routed directly to ClaimCenter review queue
- Zero-touch for weather/bandh events. Structured review for manual claim types.

BillingCenter

- Weekly premium deduction via BillingCenter direct debit scheduling
- Payout disbursement coordinated through BillingCenter's payment gateway
- Worker wallet reconciliation synced weekly

Guidewire Marketplace

- ShieldGig packaged as a Marketplace integration — any insurer on PolicyCenter/ClaimCenter can onboard ShieldGig's parametric trigger engine as a configurable product extension

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📊 Parametric Logic — Core Principle

ShieldGig does **not** calculate actual income loss. No investigation needed for automated triggers.

- A measurable disruption index is monitored in real time
- When it crosses a threshold AND falls within the worker's shift window → payout fires
- Payout = `avg_hourly_earnings × disruption_hours × 0.75`

```
Example:
  Ravi's 30-day avg earnings: ₹750/day ÷ 10 active hours = ₹75/hr
  Heavy rain confirmed: 3 hours above 64.5mm threshold
  Shift window check: disruption falls within 10 AM–10 PM  →  PASS

  Payout = ₹75 × 3 × 0.75 = ₹168.75  →  auto-disbursed to UPI
```

Why 0.75 and not 1.0 — Basis Risk: Parametric insurance by design does not perfectly match individual loss. A worker who had planned to take the day off but still had a policy active receives 75% of their average hourly earnings — a small windfall. This is basis risk, and it is intentional. The 0.75 factor prices this leakage in. Attempting to verify individual intent defeats the zero-touch model. The loss ratio target (< 0.70) accounts for this.

👤 Persona & Scenarios

Primary Persona: Ravi, Zomato Food Delivery Partner (Chennai)

Attribute	Value
Platform	Zomato (primary), Swiggy (secondary)
Weekly Earnings	₹5,200 (~₹750/day, ~₹75/hr over 10-hr shift)
Shift Window	10 AM – 10 PM derived from 30-day activity history
Peak Slots	Lunch 12–3:30 PM · Dinner 7–11:30 PM
Device	Android ~₹10,000 budget phone

Attribute	Value
Annual exposure	~80 rain days · loses ₹400–₹600/heavy rain day

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📊 Real Scenario Simulation — Chennai, November Rain

```
Date:         November 12, 2025
Location:     Adyar, Chennai
IMD data:     72mm rainfall — threshold crossed for 3 hours
Shift window: Disruption 11 AM–2 PM falls within Ravi's 10 AM–10 PM  →  PASS

Calculation:
  avg_hourly_earnings = ₹75/hr  (30-day rolling)
  disruption_hours    = 3
  coverage_factor     = 0.75
  Payout              = ₹75 × 3 × 0.75 = ₹168.75

Timeline:
  11:00 AM  →  IMD threshold crossed in Adyar
  11:02 AM  →  Shift window check: PASS
  11:02 AM  →  Fraud engine: FRS = 14/100  →  CLEAN
  11:02 AM  →  Claim pushed to Guidewire ClaimCenter
  11:04 AM  →  ₹118.13 (70%) credited to Ravi's UPI
  48 hrs    →  ₹50.62  (30%) released after review window

Total: ₹168.75.  Premium paid: ₹49.  Net benefit: ₹119.75.
```

Scenario B — Bandh / Strike (Automated)

```
NLP scraper detects: "Chennai bandh in zones 4, 7, 11"  →  confidence 0.82
Platform API: Zomato OFFLINE in zones 4, 7, 11  →  dual confirmation PASS
Shift window: bandh falls within workers' logged shift hours  →  PASS

Workers on Standard Shield:
  →  Payout = avg_hourly_earnings × disruption_hours × 0.75
  →  Auto-claim pushed to ClaimCenter
  →  Zero worker action required
```

Scenario C — Traffic Accident Blocking Zone (Manual Claim)

```
A major road accident blocks Ravi's primary delivery corridor for 2 hours.
No parametric trigger exists for this event type.
```

```
Ravi taps "Report Disruption" in the app.

What ShieldGig requires from Ravi:
  1. GPS location screenshot showing his position at time of disruption
  2. One photo or video showing the blocked road / accident scene
     (timestamped by the app at capture — EXIF data verified)
  3. Platform earnings screenshot showing zero orders in that 2-hour window

What ShieldGig cross-checks automatically:
  → Google Maps Traffic API — was speed < 5 km/h in that corridor?
  → News API — any accident or road closure reported in that area?
  → Platform order data — was order density in that zone unusually low?

If 2 of 3 cross-checks confirm:
  → Claim routed to ClaimCenter as "Assisted Manual"
  → Review SLA: 4 hours
  → Payout = avg_hourly_earnings × blocked_hours × 0.75
  → Worker notified via push + SMS with reason for outcome
```

Scenario D — Platform App Outage (Automated via Order Failure Rate)

```
Zomato status page shows "operational" but orders are failing en masse.

ShieldGig detects:
  order_failure_rate = (orders_assigned − orders_completed) / orders_assigned
  Zone failure rate  = 78%  →  threshold 60% crossed

Platform status API says UP. Order failure rate says DOWN.
Order failure rate wins — it reflects ground reality.

Workers on Standard Shield in affected zones:
  → Shift window check applied
  → Payout = avg_hourly_earnings × outage_hours × 0.75
  → Auto-claim pushed to ClaimCenter
```

🔄 End-to-End Workflow

```
mermaid
```

```
graph TD
    A[Worker opens ShieldGig] --> B[OTP login]
    B --> C[Onboarding: platform + zone + avg income]
    C --> D[ISS calculated + city risk profile applied]
    D --> E[Weekly premium quoted — bounded 0.7x–2.0x base]
    E --> F{Worker activates?}
    F -- Yes --> G[Premium via BillingCenter / UPI Autopay]
    F -- No --> H[Push nudge next morning]
    G --> I[Policy active 7 days]

    I --> J{Disruption type?}
    J -- Parametric: rain/cyclone/bandh/heat --> K[Auto trigger pipeline]
    J -- Manual: accident/road block --> L[Worker submits proof package]

    K --> M[Shift window eligibility check]
    M -- Outside shift hours --> N[No payout — not working]
    M -- Within shift hours --> O[Dual-source trigger confirmation]
    O --> P[Hourly payout calculated with 0.75 factor]
    P --> Q[Daily cap ₹300 + weekly cap ₹1000 applied]
    Q --> R[Fraud engine: FRS computed]

    L --> S[App captures GPS + photo + platform screenshot]
    S --> T[Cross-check: Traffic API + News API + order density]
    T --> U[Routed to ClaimCenter — 4hr review SLA]

    R --> V{FRS routing}
    V -- 0-30 CLEAN --> W[Auto-claim to ClaimCenter]
    V -- 31-60 REVIEW --> X[40% provisional + EXIF liveness check]
    V -- 61-100 FLAGGED --> Y[Manual queue + auto-explanation generated]

    W --> Z[70% UPI immediate + 30% after 48hr]
    X --> AA{Liveness photo verified?}
    AA -- Yes --> Z
    AA -- No --> BB[Clawback + suspension]
    Y --> CC[Worker receives which signals triggered flag + one-tap appeal]
```

💰 Weekly Premium Model

Dynamic Premium Formula

```
Weekly Premium = Base Rate × ISS Multiplier × City Risk Profile × Forecast Multiplier

Hard bounds: min(0.7x base rate) and max(2.0x base rate)
```

```
No worker ever pays less than 70% or more than 200% of their tier base rate
regardless of how extreme their risk score or forecast is.
```

ISS Multiplier:

```
ISS 80-100  →  0.85x
ISS 50-79   →  1.00x
ISS 30-49   →  1.25x
ISS 0-29    →  1.50x
```

Sunday Night Forecast Repricing: Every Sunday 11 PM the system recalculates next week's premium from the 7-day IMD forecast. Cyclone in forecast = premium rises. Clear week = premium falls. Adjustment capped at ±40% from base. Final premium then bounded by the 0.7x–2.0x hard limits.

Examples:

- Velachery worker, July peak monsoon, ISS 55 → ₹79 (near 2x cap, bounded)
- Anna Nagar worker, January dry season, ISS 82 → ₹29 (0.7x floor applied)
- Adyar worker, standard week, ISS 68 → ₹49 (standard)

Plan Tiers

Plan	Base Weekly Premium	Triggers Covered	Hourly Rate
Basic Shield	₹29	Heavy Rain	₹75/hr × 0.75
Standard Shield	₹49	Rain + Platform Downtime + Bandh	₹100/hr × 0.75
Full Shield	₹79	Rain + Platform + Bandh + Heat + Pollution + Manual	₹100/hr × 0.75
Elite Shield	₹109	All triggers + Cyclone + Manual	₹150/hr × 0.75 (cyclone)

> Scope enforced strictly: Income loss only. Health, accident, vehicle repair, and life coverage permanently excluded.

⚡ Claim Types — Automated vs Manual

Automated Parametric (zero worker action)

Trigger	Source	Fallback	Threshold	Shift Window Required
Heavy Rain	IMD station	OpenWeatherMap	⩾ 64.5 mm/day	Yes
Extreme Rain	IMD	Satellite	⩾ 115.6 mm/day	Yes
Platform Downtime	Order failure rate > 60%	Status API	> 3 hrs in zone	Yes
Bandh / Strike	NLP scraper + LLM parse	—	Confidence ⩾ 0.6 + platform offline	Yes
VVIP Gridlock	Traffic API	Rider GPS polygon	Speed < 5 km/h, 45 min	Yes
Severe Heat	IMD	OpenWeatherMap	⩾ 43°C, 2+ hrs	Yes
Severe Pollution	AQICN	State PCB	AQI ⩾ 200	Yes
Cyclone	IMD Red Alert	NDMA advisory	Official alert issued	Yes

Manual Claims (worker submits proof)

These events are real income disruptions but cannot be measured by a single external API. Worker submits a structured proof package via the app. ShieldGig cross-checks automatically and routes to ClaimCenter for a 4-hour review.

Event Type	What Worker Submits	What ShieldGig Cross-Checks	SLA
Road accident blocking zone	GPS location + timestamped photo of blockage + zero-order earnings screenshot	Google Maps speed data + News API for accident reports + zone order density drop	4 hrs
App/GPS crash during shift	Screen recording or screenshot of error + platform support ticket number	Platform API for reported incidents + device error log hash	4 hrs
Police/local authority road closure	Photo of barricade + officer notice if available	Traffic API + News API + IMD/NDMA for any related event	4 hrs
Flash mob / unannounced event	Photo/video of crowd blocking access + zone GPS timestamp	News API + social signal scraper + zone order density	6 hrs

Payout formula for manual claims is identical to automated: `avg_hourly_earnings × verified_disruption_hours × 0.75`

The difference is human-in-the-loop verification replaces the automated trigger. Payout value is the same.

🔒 Shift Window Eligibility Check

A claim only fires if the disruption falls within the worker's established working hours. Derived from their 30-day delivery activity history — not self-declared.

python

```python
def shift_window_check(worker, disruption_start, disruption_end):
    # Worker's typical shift window inferred from 30-day GPS + order history
    typical_start = worker.activity_graph.median_start_hour  # e.g. 10
    typical_end   = worker.activity_graph.median_end_hour    # e.g. 22

    disruption_overlap_hours = calculate_overlap(
        disruption_start, disruption_end,
        typical_start, typical_end
    )

    if disruption_overlap_hours == 0:
        return False, 0   # Disruption outside shift window — no payout

    return True, disruption_overlap_hours  # Pay for overlapping hours only
```

Why this matters:

- Rain at 2 AM does not trigger a payout for a worker who sleeps at 2 AM
- Reduces basis risk (workers who planned days off can't claim full events)
- Reduces fraud surface (spoofing a flood at 3 AM when you'd never be working anyway is now worthless)
- New workers (< 2 weeks) default to a conservative 9 AM–9 PM window until history builds

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💧 Liquidity Protection Caps

```
Per disruption hour  →  ₹75–₹150 × 0.75 coverage factor
Daily cap            →  ₹300/day regardless of hours or trigger count
Weekly cap           →  ₹1,000/week hard ceiling per worker
Pool reserve         →  25% of weekly premium pool held back
                        Released only if claims exceed 70% of pool
Loss ratio target    →  < 0.70
```

Maximum weekly liability per worker is known before the week starts — not discovered after a disaster.

🤖 AI/ML — Five Models

Model 1: Income Stability Score (ISS)

Scores every worker 0–100 at onboarding. Determines premium multiplier.

Real datasets used:

- IMD District Rainfall 2015–2024 — imdpune.gov.in — zone-level heavy rain probability per pin-code
- PLFS Gig Worker Earnings 2023 — mospi.gov.in — real earnings distributions for premium calibration
- data.gov.in Pincode Directory — maps worker pin-code to IMD district zone risk score

Phase 1 — Rule engine:

```python
```

```python
def calculate_iss(zone_flood_risk, avg_daily_income,
                  disruption_freq_12mo, claims_history_penalty):
    score = 100
    score -= zone_flood_risk * 20
    score -= min(disruption_freq_12mo, 15)
    score += min(avg_daily_income / 200, 10)
    score -= claims_history_penalty
    return max(0, min(100, score))
```

Phase 2: XGBoost. Adopted only when precision exceeds rule engine baseline by > 5% on holdout set.

Model 2: Sunday Night Premium Repricing

Fetches 7-day IMD forecast every Sunday 11 PM. Recalculates premium before the disruption, not after. Final premium bounded by 0.7x–2.0x hard limits regardless of forecast risk.

Model 3: Fraud Detection Engine (FRS 0–100)

Real datasets used:

- Kaggle Auto Insurance Claims — 15,000+ labeled fraud/non-fraud claims for supervised baseline
- Live GPS + OpenCelliD cell tower signals — real-time location authenticity
- Worker behavioral history — internal DB — delivery heatmaps, shift windows, claim history

Six signal layers:

Layer 1 — Location Authenticity

Signal	What It Detects
Cell tower triangulation (OpenCelliD)	Physical location independent of GPS
IP geolocation (MaxMind)	Home broadband vs mobile data in field; VPN/proxy detection
Wi-Fi SSID fingerprint	Home Wi-Fi present during claimed outdoor disruption
GPS jitter analysis	Spoofed GPS has statistically perfect coordinates — real GPS micro-drifts naturally over a 5-minute window

A spoofer at home shows: perfect GPS coordinates (zero variance) + home SSID + home broadband IP + no jitter. Four contradictions in one claim.

Layer 2 — Shift Window Behavioral Baseline Claim checked against Personal Activity Graph. Zone never previously worked in = heavy FRS penalty.

Layer 3 — Claim Initiation Latency Claims filed < 30 seconds after a trigger fires are flagged. Genuine workers don't have push-to-claim reflexes. Syndicate members watching a Telegram channel do.

Layer 4 — Coordinated Ring Detection

```
If N workers claim the same grid cell within 15 minutes
AND onboarding dates / device subnets / referral chains are correlated:
  → Entire cluster flagged
  → Isolation Forest: anomaly score > 3σ from historical claim velocity
  → Poisson distribution test on claim timing:
      Real disruptions = gradual filing over 20–40 min
      Coordinated rings = uniform firing within seconds
      Uniform distribution at p < 0.05 = ring signal confirmed
```

Layer 5 — Weather Cross-Verification At least one IMD station within 5km must independently confirm the trigger. No corroborating ground station = flagged.

Layer 6 — Worker Trust Score *(backend only — not visible in UI)* Workers with 8+ weeks of clean claim history receive a Trust Score that reduces their effective FRS by up to 15 points. Long-tenure honest workers are actively protected from false positives without knowing the mechanism exists.

FRS routing:

```
0–30   CLEAN    →  auto-approve via ClaimCenter
31–60  REVIEW   →  40% provisional payout + EXIF liveness check
61–100 FLAGGED  →  manual review queue + auto-explanation generated
```

Model 4: NLP Disruption Scraper + LLM Preprocessing

Phase 1 — spaCy keyword scoring: Polls news feeds every 15 min for bandh/strike/curfew terms. Scores confidence per article. Requires platform API to confirm deliveries offline before trigger fires.

Phase 2 — LLM preprocessing layer: An LLM (lightweight, restricted role) converts unstructured IMD bulletins, government advisories, and local incident reports into structured JSON signal objects consumed by the deterministic rule engine. The LLM touches preprocessing only — it never participates in the YES/NO payout decision. Every payout decision is deterministic, auditable, and explainable to regulators.

```
Unstructured input:
  "IMD issues orange alert for heavy to very heavy rainfall in Chennai
   districts on November 12-13 due to low pressure system in Bay of Bengal"

LLM output (structured JSON):
  {
    "alert_type": "heavy_rainfall",
    "severity": "orange",
    "zones": ["Chennai"],
    "start_date": "2025-11-12",
    "end_date": "2025-11-13",
    "source": "IMD",
    "confidence": 0.96
  }

Rule engine receives JSON → checks threshold → fires trigger or not.
```

Model 5: Facebook Prophet — Disruption Forecasting *(Phase 3)*

Forecasts next 4 weeks of disruption frequency per zone and projected loss ratio. Feeds the insurer admin dashboard with forward-looking capital reservation estimates.

Real datasets used:

- IMD District Rainfall 2015–2024 — seasonal disruption frequency baseline
- WAQI Historical AQI — aqicn.org — pollution event frequency per city
- Internal claim + payout data accumulated from platform operation

Why Prophet: Built-in seasonality decomposition handles India's monsoon cycle (weeks 24–40 are structurally high-frequency) without manual feature engineering. Handles data gaps gracefully — critical for a startup with sparse early-stage data. Gives insurers accurate forward-looking loss estimates for capital reservation.

Output: Next 4 weeks of expected disruption events per zone + predicted loss ratio. Loss ratio target: < 0.70. If forecast exceeds 0.70, premium repricing and pool reserve alerts fire automatically.

🛡️ Adversarial Defense & Anti-Spoofing Strategy

> *Response to DEVTrails 2026 Market Crash: 500-worker syndicate used GPS-spoofing apps to drain the liquidity pool.*

Core Principle

A real crisis leaves a consistent fingerprint across multiple independent signals. A spoofer can only fake one or two. ShieldGig exploits this asymmetry.

What the Syndicate Cannot Fake

1. Cell tower IDs consistent with the claimed physical zone (OpenCelliD)
2. Natural GPS micro-drift — spoofed coordinates have zero statistical variance
3. Mobile data IP in field — home broadband shows up via MaxMind geolocation
4. Official IMD/NDMA advisories for the claimed crisis
5. 30-day delivery heatmap showing they operate in the claimed zone
6. Claim timing variance — rings fire uniformly; real disruptions spread over 20–40 min (Poisson test)
7. Independent onboarding paths — 500 members collapse to a small referral/device graph

Protecting Honest Workers — EXIF Liveness + Transparent Appeals

For borderline claims (FRS 31-60):

```
→  40% provisional payout released immediately
→  Worker receives one-tap camera prompt
→  Live photo captured — app reads embedded GPS EXIF data
→  EXIF coordinates match claimed zone?
     YES  →  remaining 60% released instantly
     NO   →  provisional 40% clawed back, account suspended
```

For flagged claims (FRS 61–100):

```
→  Claim held for manual review
→  Auto-explanation generated immediately:
     "Your claim was flagged because:
      - Your device was connected to a Wi-Fi network
        not associated with your operating zone
      - GPS coordinates showed unusually low variance
        during the claimed disruption window
      You can appeal this decision."
→  One-tap appeal pathway in app
→  Human review with AI summary: 4-hour SLA
→  First-time flags: enhanced monitoring only
→  Confirmed multi-signal fraud: suspension with defined dispute process
→  No permanent blacklisting without human review
```

Zone context override: When IMD, NDMA, or government sources confirm a declared disaster, zone-level fraud thresholds are automatically elevated. During officially declared emergencies, the system assumes good faith and shifts burden of proof.

📈 Income Protection Dashboard

Worker View

```
This Week
  Premium paid:       ₹49
  Payout received:    ₹168.75  (3-hr heavy rain, 0.75 factor applied)
  Net protected:      ₹119.75  ✅

  "Without ShieldGig this week, you would have lost ~₹225."

Last 4 weeks:
  Wk 1: paid ₹49  received ₹0        clear week
  Wk 2: paid ₹49  received ₹168.75   3-hr heavy rain
  Wk 3: paid ₹49  received ₹112.50   2-hr platform downtime
  Wk 4: paid ₹49  received ₹0        clear week
```

Insurer / Admin View

- Live: active policies, premium pool, loss ratio vs 0.70 target, fraud flagged count
- 72-hour payout forecast + Prophet 4-week loss ratio projection
- Pool reserve status: current % vs 25% floor
- Fraud review queue → synced to ClaimCenter with auto-explanation summaries
- Zone risk heatmap by city-specific disruption profile
- Weekly: premium collected vs payout released (8-week trend)

⚠️ System Reliability

Failure Mode	Handling
OpenWeatherMap unavailable	Fall back to IMD station feed
IMD data delayed	Last-confirmed reading + 30-min cache
Platform API unreachable	Order failure rate used as primary signal
GPS signal lost on device	Last verified location within 15-min window

Failure Mode	Handling
NLP scraper fails	Trigger held; manual admin review
Single-source trigger only	Held for admin confirmation — no auto-payout
MaxMind IP API unavailable	Wi-Fi fingerprint signal weighted up; IP layer skipped

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🏗 Platform Decision — Mobile PWA (Flutter)

Delivery workers do not use laptops. Every interaction happens on a ₹10,000 Android phone at a red light. Designed for one-thumb operation, 3-second tasks, push + SMS as primary engagement channels.

Anti-spoofing engine requires direct access to device signals — cell tower IDs, Wi-Fi SSID fingerprints, GPS jitter analysis. PWAs cannot reliably access these. Flutter provides full native sensor access plus a single codebase for both Android (worker app) and web (insurer admin dashboard).

🛠 Tech Stack

Frontend

Component	Technology
Framework	Flutter (Dart) — PWA + Web
State Management	Riverpod
Background Location	flutter_background_geolocation
Local Storage	Hive (offline-first)
Payments (mock)	Razorpay Flutter SDK — test mode
Notifications	Firebase Cloud Messaging + Twilio SMS fallback

Backend

Component	Technology
API Server	Node.js + Express
Database	Supabase (PostgreSQL + PostGIS)
Auth	Supabase Auth (OTP via phone)
Hosting	Render (free tier)
Trigger Polling	Node-cron (every 15 min)
NLP Scraper	Python + spaCy via FastAPI microservice

AI/ML

Component	Technology
ISS Scoring (Phase 1)	Python rule engine via FastAPI
Premium Repricing	Forecast-weighted rule engine → XGBoost Phase 2
Fraud Detection	scikit-learn Isolation Forest + deterministic rule layers
NLP Scraper	spaCy Phase 1 → LLM preprocessing Phase 2
Disruption Forecasting	Facebook Prophet — Phase 3

Guidewire

Integration	API
Policy lifecycle	PolicyCenter REST API
Claim creation + routing (auto + manual)	ClaimCenter REST API
Premium billing + payout	BillingCenter REST API
Distribution packaging	Guidewire Marketplace

External APIs

API	Use	Cost
OpenWeatherMap	Rainfall real-time	Free (1,000 calls/day)
IMD Open Data	Authoritative thresholds + fallback	Free
AQICN / WAQI	AQI monitoring	Free token
MaxMind GeoIP2	IP geolocation + VPN/proxy detection	Free tier
OpenCelliD	Cell tower triangulation	Free tier
Google Maps Traffic	Road speed for gridlock + accident cross-check	Pay-per-use
Brave Search + NewsAPI	Crisis event corroboration for manual claims	Free tier
Zomato/Swiggy	Order failure rate + platform status	Mock Phase 1
Razorpay	UPI payout simulation	Test mode

🧪 MVP Scope — Phase 1

Proving the parametric loop. Not an enterprise system.

Phase 1 demonstrates:

- Rain trigger via live OpenWeatherMap + IMD with shift window check
- Personalized payout using 30-day rolling avg earnings × 0.75 factor
- NLP scraper for bandh detection (mock news feed)
- Hourly payout with daily ₹300 + weekly ₹1,000 caps
- ISS scoring (rule engine) with named real datasets
- Fraud: GPS jitter + cell tower + IP geolocation + claim velocity
- EXIF liveness check flow for borderline claims
- Auto-explanation generation for flagged claims
- Manual claim submission flow (UI + proof capture)
- Order failure rate trigger for platform outage
- Guidewire ClaimCenter payload structure (auto + manual variants)
- UPI payout via Razorpay test mode

Phase 2 adds: full Flutter app · live Guidewire integration · LLM news preprocessing · city risk profiles

Phase 3 adds: Prophet forecasting model · Isolation Forest fraud model · admin dashboard · Marketplace packaging

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💸 Cost Efficiency

Resource	Cost
OpenWeatherMap, IMD, AQICN	₹0
MaxMind GeoIP2	₹0 (free tier)
OpenCelliD	₹0 (free tier)
Brave Search + NewsAPI	₹0 (free tiers)
Supabase + Render	₹0 (free tiers)
Razorpay test mode	₹0

Total infrastructure: ₹0/month.

📅 6-Week Plan

✅ Phase 1 (Weeks 1–2) — Current

- [x] Shift window eligibility architecture
- [x] Personalized payout formula with 0.75 coverage factor
- [x] Hourly payout + daily/weekly caps
- [x] GPS jitter + IP geolocation fraud layers
- [x] Poisson distribution ring detection
- [x] NLP scraper + LLM preprocessing architecture
- [x] Named real datasets identified for all models
- [x] Manual claim types designed with proof requirements
- [x] Platform outage via order failure rate
- [x] Transparent appeals + auto-explanation system
- [x] Premium bounds (0.7x–2.0x)
- [x] Worker Trust Score (backend design)
- [x] Prophet forecasting plan (Phase 3)
- [x] City risk profiles designed
- [x] Guidewire integration mapped
- [ ] Flutter scaffold + Supabase schema

- [ ] ISS rule engine (Python)
- [ ] Phase 1 demo video

Phase 2 (Weeks 3-4) — Automation & Protection

- [ ] Full Flutter app — all 5 screens + manual claim flow
- [ ] Weather + NLP trigger cron live
- [ ] Order failure rate trigger for platform outage
- [ ] MaxMind + OpenCelliD integration
- [ ] GPS jitter analysis in fraud engine
- [ ] EXIF liveness check flow
- [ ] Auto-explanation generation for rejections
- [ ] Live ClaimCenter/PolicyCenter integration
- [ ] City risk profiles: Chennai + Mumbai + Bengaluru + Kolkata

Phase 3 (Weeks 5-6) — Scale & Optimise

- [ ] Isolation Forest fraud model + Poisson timing test
- [ ] LLM news preprocessing pipeline
- [ ] Facebook Prophet forecasting model
- [ ] Insurer admin dashboard + Prophet projections
- [ ] Pool reserve monitor
- [ ] Worker Trust Score accumulation logic
- [ ] Guidewire Marketplace packaging
- [ ] Final 5-min demo video

📊 Business Viability

Metric	Value
Target workers — Chennai pilot	10,000
Average weekly premium	₹49
Weekly premium pool	₹4,90,000
Pool reserve (25%)	₹1,22,500 held
Loss ratio target	< 0.70
Estimated real loss ratio (capped + 0.75 factor)	~45-55%
Traditional claims processing cost	₹1,000-₹3,000/claim
ShieldGig automated claims cost	₹0

Metric	Value
ShieldGig manual claims cost	~₹200 (4-hr human review allocation)

🤝 IRDAI Compliance

- Technology partner model — not a licensed insurer
- Policy under partner insurer's IRDAI license
- Triggers rely on IMD — IRDAI-recognized data source
- Payout terms transparent at activation (parametric requirement)
- Microinsurance compliant: ₹29–₹109/week, simplified format
- Within IRDAI Regulatory Sandbox guidelines for parametric products (2019)

👥 Team

Member	Role
[Name 1]	Flutter Development
[Name 2]	Backend / API + Guidewire Integration
[Name 3]	AI/ML + Fraud Engine + NLP + Prophet
[Name 4]	UI/UX Design
[Name 5]	Insurance Domain + City Risk Profiles + Pitch

🎬 Phase 1 Deliverables

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ShieldGig — Real-time income protection for the workers who keep our cities fed.